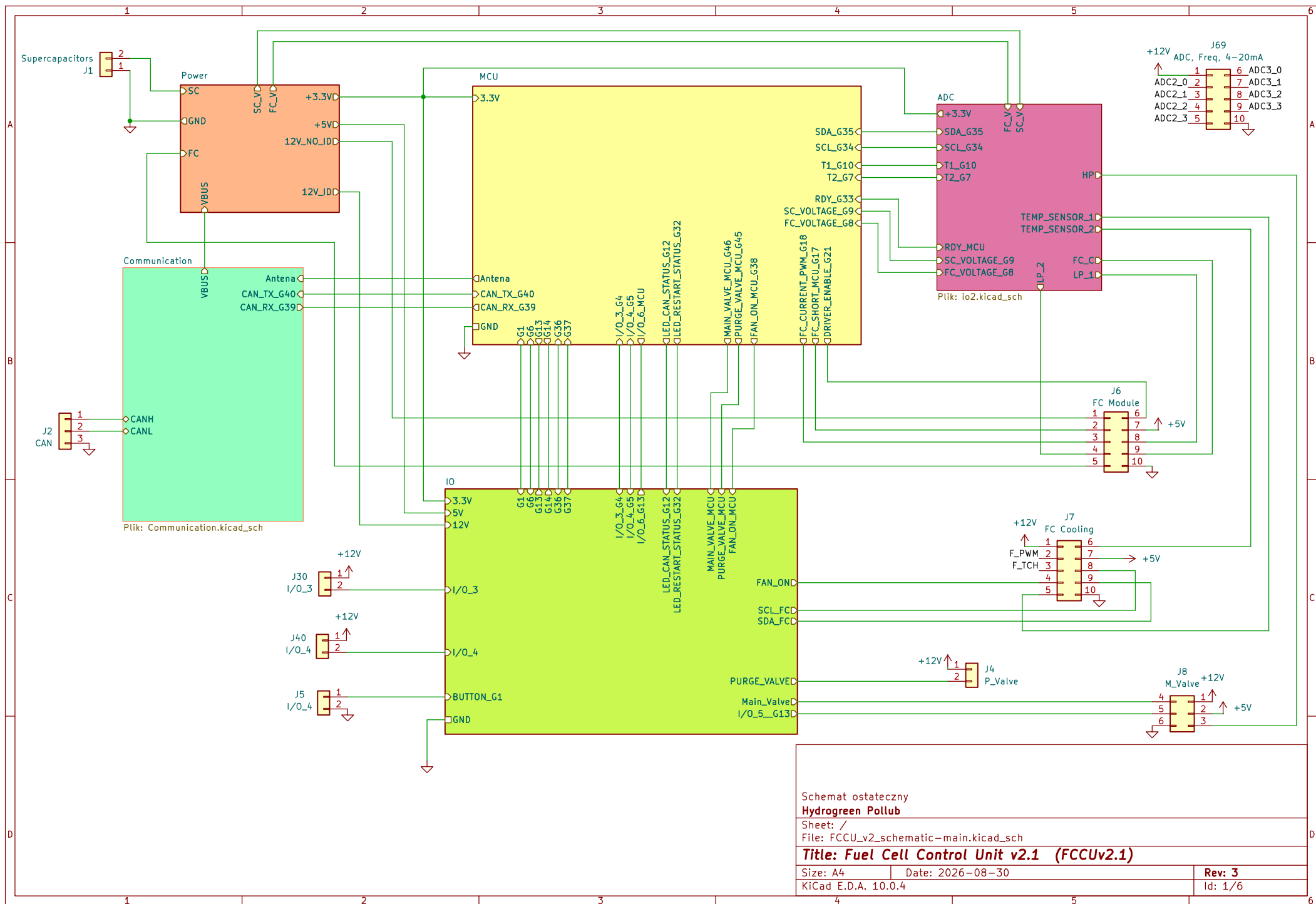




Schemat ostateczny

Hydrogreen Pollub

Sheet: /

File: FCCU_v2_schematic-main.kicad_sch

Title: Fuel Cell Control Unit v2.1 (FCCUv2.1)

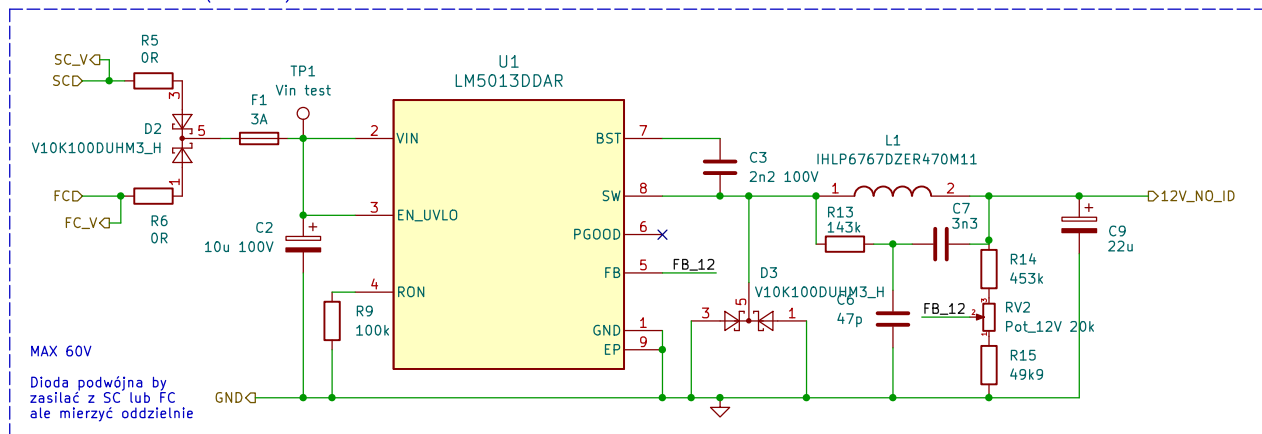
Size: A4 Date: 2026-08-30

KiCad E.D.A. 10.0.4

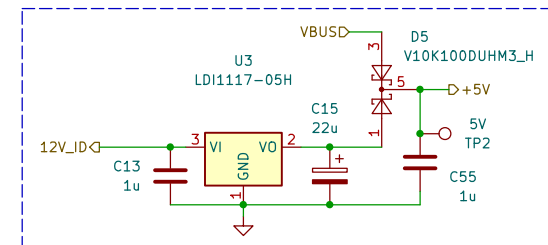
Rev: 3

Id: 1/6

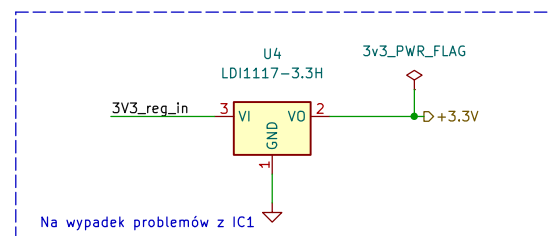
Buck converter (+12 V)



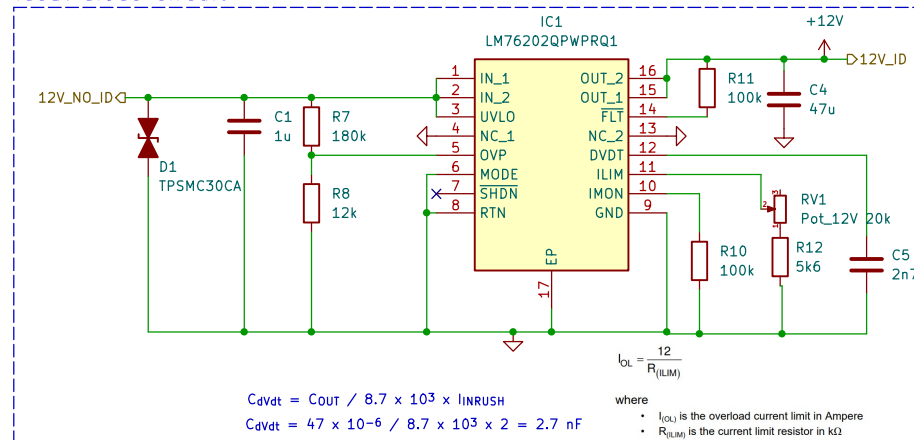
Simple LDO (+5 V)



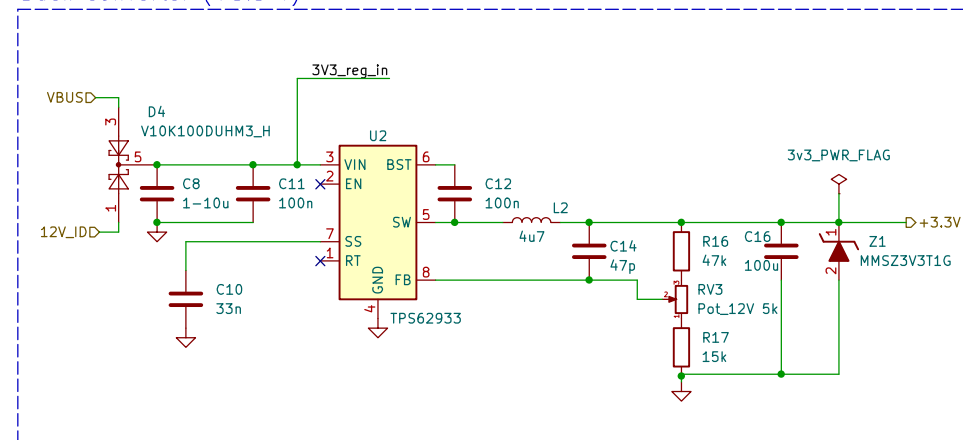
Additional LDO (+3.3 V)



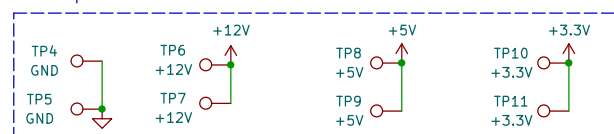
Ideal diode circuit



Buck converter (+3.3 V)



Test points



Schemat ostateczny

Hydrogreen Pollub

Sheet: /Power/

File: schematic-power.kicad_sch

Title: Fuel Cell Control Unit v2.1 (FCCUv2.1)

Size: A4

Date: 2026-08-30

KiCad E.D.A. 10.0.4

Rev: 3

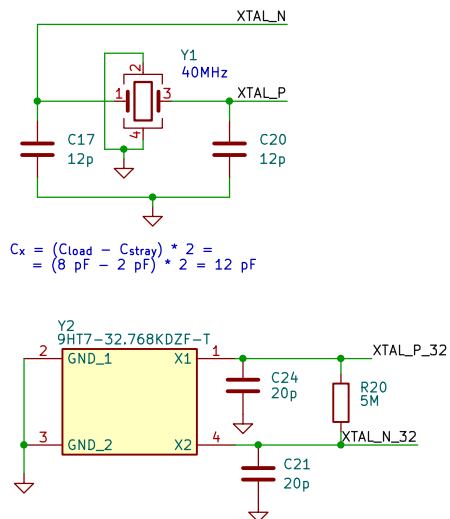
Id: 2/6

Microcontroller

U6
ESP32-S3FN8

ESP32-S3
56pin

Crystal oscillator



$$C_x = (C_{load} - C_{stray}) * 2 = (8 \text{ pF} - 2 \text{ pF}) * 2 = 12 \text{ pF}$$

$$C_x = (C_{load} - C_{stray}) * 2 = (12.5 \text{ pF} - 2.5 \text{ pF}) * 2 = 20 \text{ pF}$$

Schemat ostateczny

Hydrogreen Pollub

Sheet: /MCU/

File: schematic-mcu.kicad_sch

Title: Fuel Cell Control Unit v2.1 (FCCUv2.1)

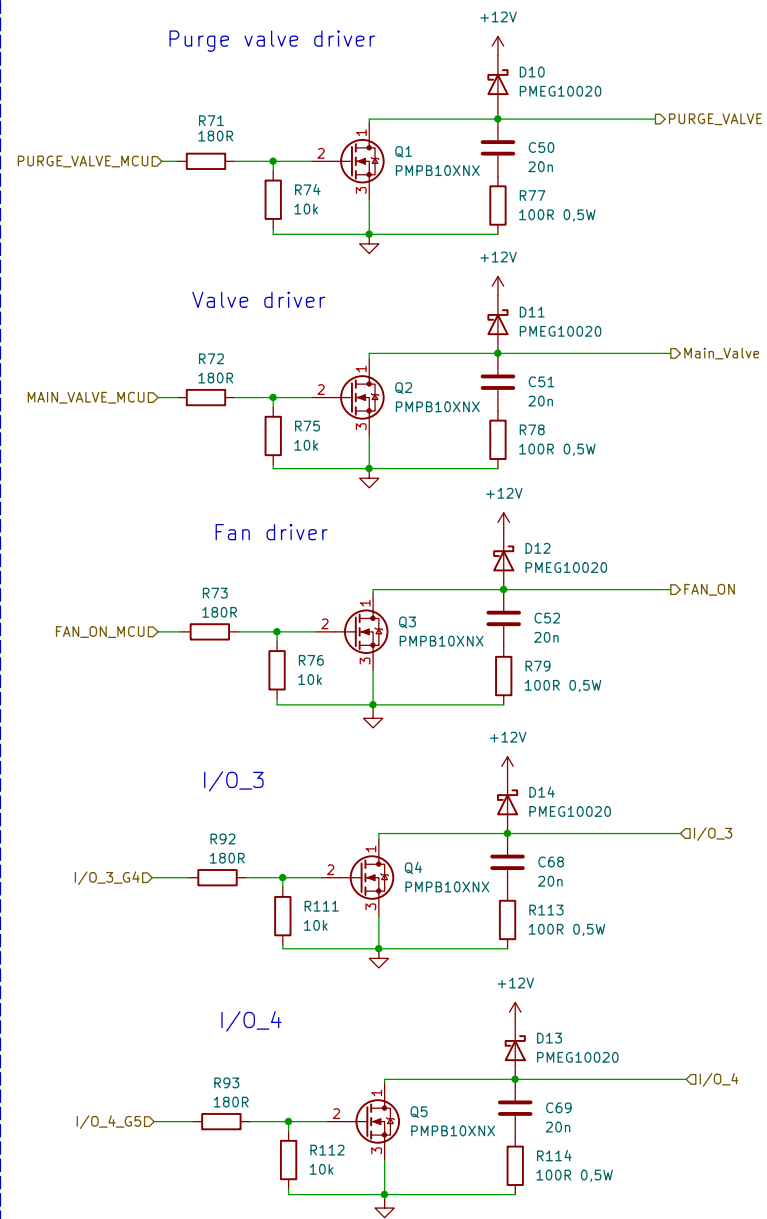
Size: A4 Date: 2026-08-30

KiCad E.D.A. 10.0.4

Rev: 3

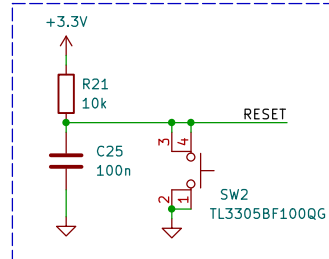
Id: 3/6

N-mosfet Outputs

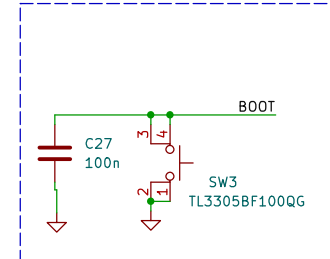


ESP32 S3 standardowo może podać 20mA (165R),
ale przy ustawieniu rejestru chwilowo 40mA => minimum 83R
Przy 180R tranzystory możnaysterować PWM 5-95% 20kHz
I/O 3 i 4 można wykorzystać do sterowania zaworem proporcjonalnym i nawilżaczem

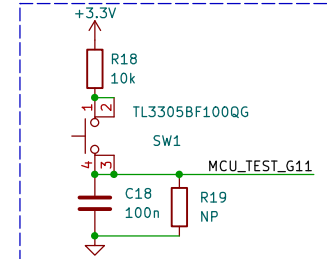
Reset button



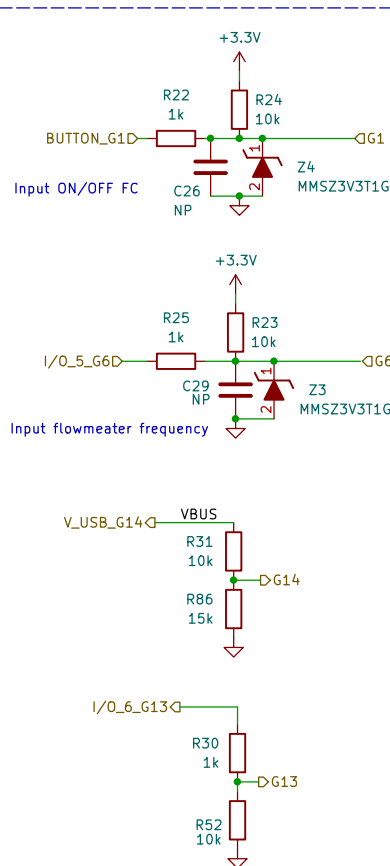
Boot button



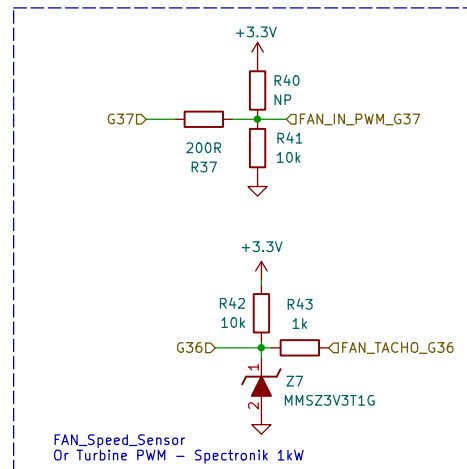
Test MCU or FC control



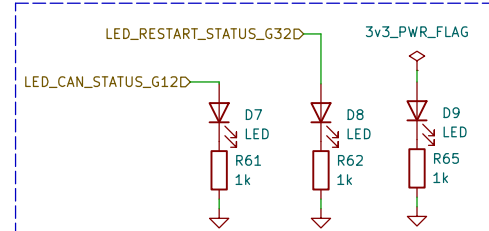
Additional mesurments



Fans PWM Control



Signal diodes



Schemat ostateczny
Hydrogreen Pollub

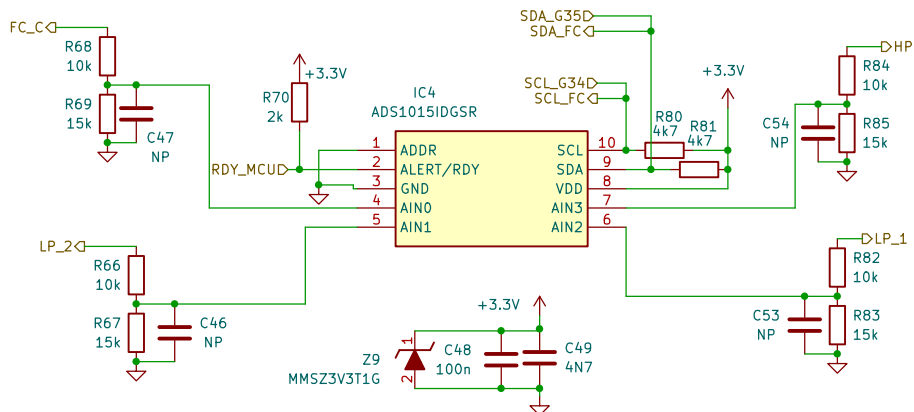
Sheet: /10/
File: schematc-io.kicad_sch

Title: Fuel Cell Control Unit v2.1 (FCCUv2.1)

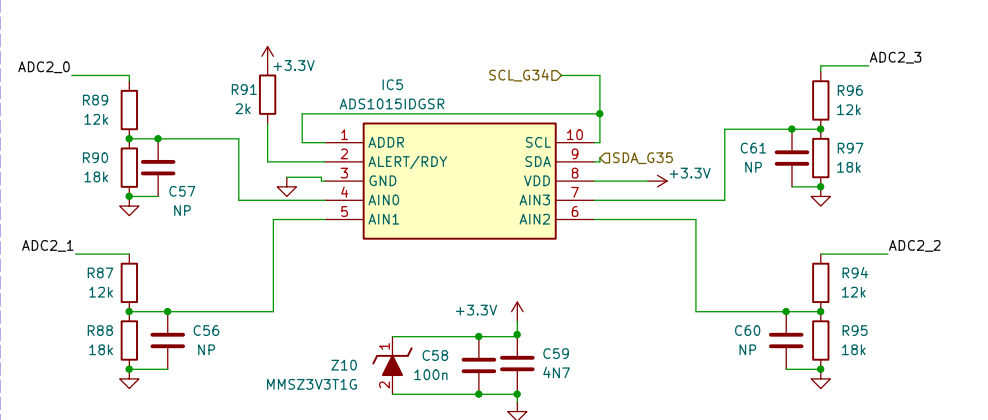
Size: A4 Date: 2026-08-30
KiCad E.D.A. 10.0.4

Rev: 3
Id: 4/6

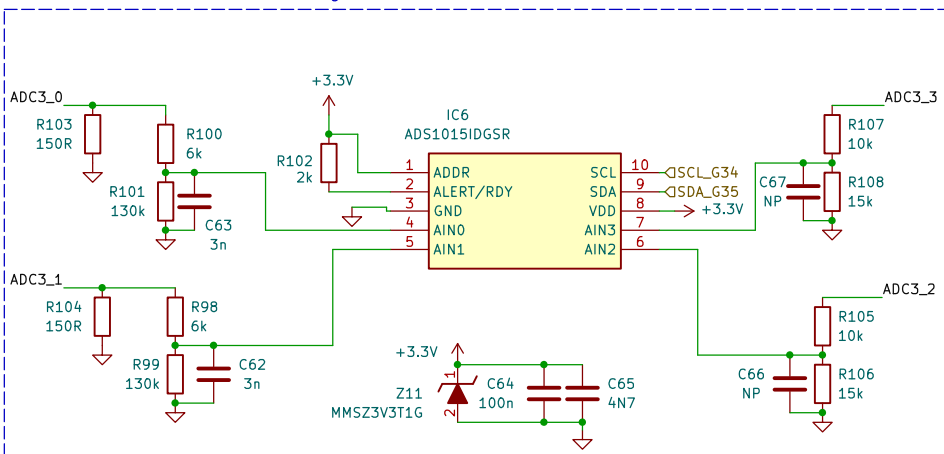
ADC1 0–5V Pressure Sensors



ADC2 0–5V Hall Current Sensors



ADC3 Current 0–20mA & Voltage 0–5V



12k i 18k lub 10k i 15k -> 5V->3V
10k i 20k -> 5V->3.33V

Dzielniki na max 2.9V AIN ESP
(napięcie zakresu powinno być maksymalnie 0.3V wyższe
od VDD, a ESP nie toleruje 5V)

NP – not populated -> nie montować bez przemyślenia (zmniejszenie dynamiki sygnału)
Można dać diody zenera otwierając się przy 3.5V (ESP32 S3 wytrzyma max 3.6V)

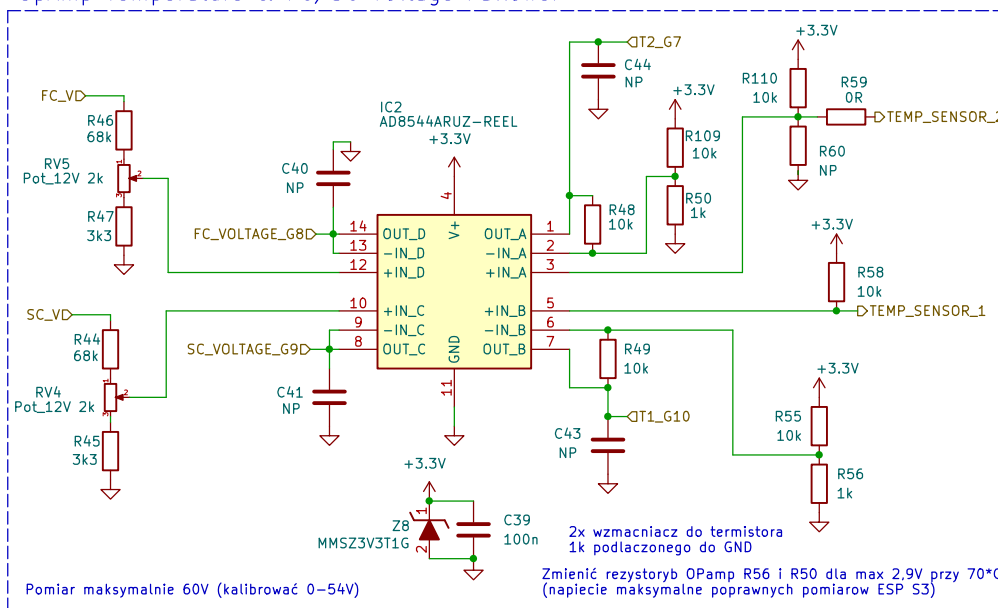
ADDR ustawia jeden z 4 adresów:

GND 0x48
VDD 0x49
SDA 0x4A
SCL 0x4B

Pullup i2c z wszystkich pytek BME i ADS nie może
obniżyć rezystancji poniżej 2k

Dodatkowy pomiar wilgotności, temperatury i ciśnienia modułem BME w ogniwie i nakładką z i2c

OpAmp Temperature & FC/SC Voltage Follower



Pomiar maksymalnie 60V (kalibrować 0–54V)

2x wzmacniacz do termistora
1k podłączonego do GND
Zmienić rezystory OpAmp R56 i R50 dla max 2.9V przy 70°C
(napięcie maksymalne poprawnych pomiarów ESP S3)

Schemat ostateczny
Hydrogreen Pollub

Sheet: /ADC/
File: io2.kicad_sch

Title: Fuel Cell Control Unit v2.1 (FCCUv2.1)

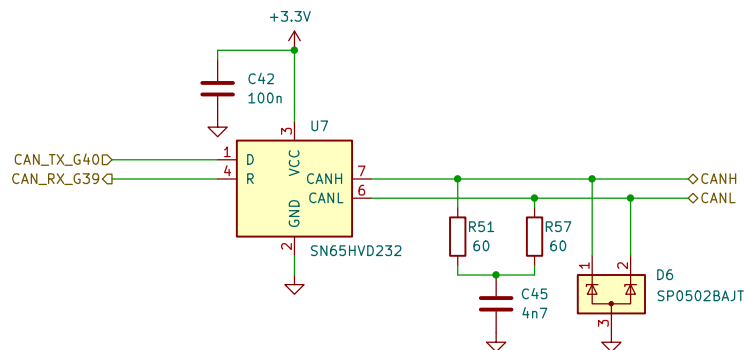
Size: A4 Date: 2026–08–30

KiCad E.D.A. 10.0.4

Rev: 3

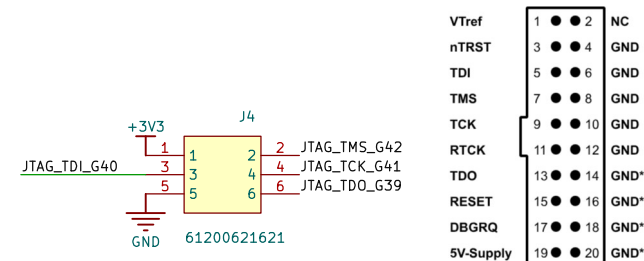
Id: 5/6

CAN Transceiver



Rezystor terminujące R51 i R57 montować tylko na końcach magistrali

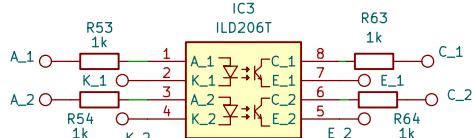
JTAG mini connector



Złącze Jtag – nie kompatybilne z adapterem Wikt
(trzeba przeciąć ścieżkę Resetu i połączyć z pinem 5 dużego złącza Jlink)

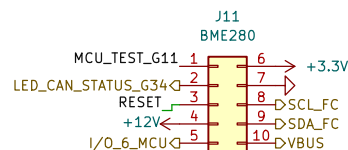
<https://www.segger.com/products/debug-probes/j-link/technology/interface-description/>

Dual transistor

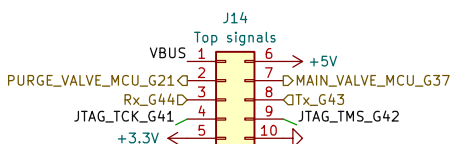
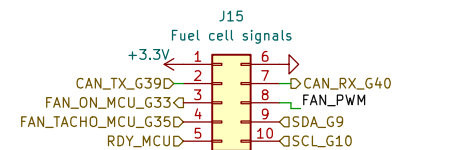
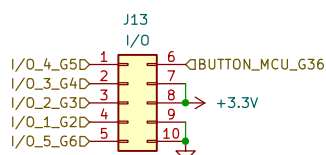


Transistor do zewnętrznego drivera regulacji prądu i zwarć
Niezbędne podłączenie do wybranych I/O
Możliwy montaż drivera zamiast IC3

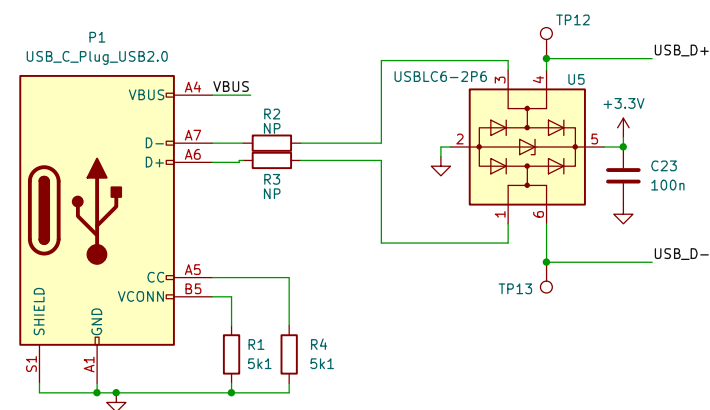
Additional Goldpins



Na J11 można wybrać pomiar
napięcia 12V lub 5V USB I/O_6
Nie montować płytek BME280 odwrotnie!

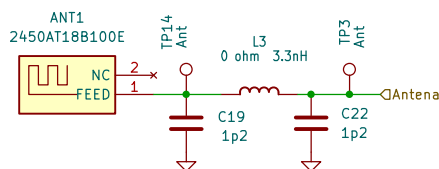


USB-C (programming and debug)



Zastosowane USB C posiada lutowane pady, a nie tylko nóżki
Błędny montaż sprawi, że złącze będzie jednostronne

Wi-Fi & Bluetooth antenna



Wartości cewki i cap – nota anteny + pomiary

Schemat ostateczny

Hydrogreen Pollub

Sheet: /Communication/

File: Communication.kicad_sch

Title: Fuel Cell Control Unit v2.1 (FCCUv2.1)

Size: A4 Date: 2026-08-30

KiCad E.D.A. 10.0.4

Rev: 3

Id: 6/6